

Do Concealed Carry Laws Reduce Crime?

A Panel Data Analysis of U.S. States (1977–1999)

1. Introduction

The relationship between gun control policies and crime remains one of the most debated topics in public policy. In particular, "shall-issue" concealed carry laws have been argued to either deter crime or increase violence. This report examines whether the adoption of such laws has a causal effect on violent crime rates in the United States.

The key challenge is distinguishing correlation from causation. States that adopt these laws may differ systematically from those that do not. Therefore, a simple comparison may lead to biased conclusions.

2. Data

The dataset consists of panel data for 51 U.S. states (including Washington D.C.) from 1977 to 1999, yielding 1,173 observations.

Key Variables

- Violent crime rate (vio)
- Murder rate (mur)
- Robbery rate (rob)
- Shall-issue law indicator (shall)
- Incarceration rate (incarc_rate)
- Socioeconomic controls:
 - Average income (avginc)
 - Population (pop)
 - Population density (density)
 - Demographic composition (pb1064, pw1064, pm1029)

3. Descriptive Statistics and Data Features

Several important features emerge from the summary statistics:

First, violent crime exhibits substantial variation, ranging from 47 to 2921 per 100,000 population, indicating strong right-skewness. To address this, the dependent variable is log-transformed in the analysis.

Second, the policy variable (shall) takes the value 1 in approximately 24% of observations. Since states adopt the law at different points in time, the dataset reflects a staggered policy adoption setting. Third, there is significant heterogeneity across states. Variables such as population and density vary widely, suggesting that unobserved state characteristics may influence crime rates.

Finally, incarceration rates differ substantially across observations, indicating that multiple policy dimensions may simultaneously affect crime.

These features motivate the use of panel data methods with fixed effects.

4. Empirical Strategy

To estimate the causal effect of shall-issue laws, this study employs a two-way fixed effects (TWFE) model:

- State fixed effects control for time-invariant differences across states
- Year fixed effects control for common shocks over time

This specification corresponds to a difference-in-differences framework.

Identification Assumption

The key assumption is parallel trends: in the absence of the policy, treated and untreated states would follow similar trends in crime rates.

Model Comparison

To assess robustness, multiple specifications are estimated:

1. Simple OLS
2. OLS with controls
3. State fixed effects
4. Two-way fixed effects

A Hausman test is conducted to compare fixed and random effects models.

5. Results

The results vary across model specifications.

Simple OLS estimates are not statistically significant, suggesting that naive correlations may be misleading.

After adding controls, the estimated effect becomes negative and statistically significant. However, this estimate may still be biased due to unobserved heterogeneity.

The preferred specification is the two-way fixed effects model, which finds that shall-issue laws are associated with approximately a 3.8% reduction in violent crime rates.

However, the magnitude is modest, and results are sensitive to model choice.

6. Robustness Checks

Several robustness checks are conducted:

- Clustered standard errors at the state level
- Alternative model specifications
- Analysis by crime type

The direction of the effect remains broadly consistent, but statistical significance varies.

7. Limitations

This analysis has several limitations:

- Policy adoption may be endogenous
- Potential time-varying omitted variables
- Parallel trends assumption is not fully tested (no event study)

These factors limit the strength of causal interpretation.

8. Conclusion

This study finds that shall-issue concealed carry laws are associated with a modest reduction in violent crime when controlling for state and time effects.

However, the results are sensitive to model specification and should be interpreted with caution. The findings highlight the importance of careful identification in policy evaluation and demonstrate that statistically significant results do not necessarily imply large practical effects.

9. Reproducibility

The analysis can be replicated using the provided code:

- Stata: run analysis.do
- Python: run analysis.py

All outputs are generated from the raw data using these scripts.
